ELECTRIC VEHICLE POPULATION ANALYSIS

An Interactive Tableau Dashboard for Washington State

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Tools: Tableau, Data Visualization, Descriptive Analytics
Dataset: Electric Vehicle Population Data

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1. Executive Summary

This project explores the electric vehicle (EV) population in Washington State using Tableau. The objective is to develop an interactive dashboard that makes it easier to understand the distribution of electric vehicles across manufacturers, models, vehicle types, and model years.

The analysis uses a structured vehicle population dataset containing information about vehicle identification, manufacturer, model, model year, electric range, and manufacturer-suggested retail price (MSRP).

A Tableau dashboard was developed to combine key performance indicators (KPIs), comparative visualizations, and detailed vehicle records. The dashboard includes manufacturer comparison, model-wise vehicle distribution, vehicle-type distribution, and a model-year analysis.

Interactive controls allow users to explore the data by model year and electric vehicle type, while a manufacturer selector highlights a selected manufacturer in the comparison chart.

The project demonstrates the practical application of business intelligence and data visualization techniques to organize complex data, identify distribution patterns, and support exploratory analysis.

2. Introduction

Electric vehicles have become an important part of the automobile industry's transition towards alternative transportation technologies. Understanding their distribution and characteristics can provide useful context for examining the development of the EV market.

Vehicle population datasets contain information about manufacturers, models, vehicle types, model years, electric range, and pricing. However, analysing these records individually can make it difficult to identify broader patterns or compare different categories effectively.

This project uses Tableau to transform electric vehicle population data into an interactive dashboard.

The dashboard focuses on the distribution of recorded vehicles across different manufacturers, models, model years, and electric vehicle types. It also presents summary metrics such as total vehicle records, average electric range, average MSRP, and the number of manufacturers.

The project is primarily descriptive and exploratory. It focuses on understanding the available data rather than predicting future vehicle demand or establishing the causes of electric vehicle adoption.

3. Problem Statement

The Electric Vehicle Population dataset contains a substantial number of vehicle records, making manual analysis and comparison difficult.

Without appropriate aggregation and visualization, it can be challenging to identify:

- Differences in vehicle counts across manufacturers.
- The distribution of electric vehicles across model years.
- The representation of Battery Electric Vehicles and Plug-in Hybrid Electric Vehicles.
- Differences in vehicle counts across individual models.
- Summary characteristics such as average electric range and MSRP.

The objective of this project is to address these analytical challenges through an interactive Tableau dashboard that presents the information in a structured and accessible manner.

4. Project Objectives

The primary objectives of the project are:

1. Summarize the electric vehicle population using key performance indicators.
2. Compare vehicle counts across manufacturers.
3. Examine the distribution of electric vehicle records across model years.
4. Analyse the representation of different electric vehicle types.
5. Compare vehicle counts across individual models.
6. Examine average electric range and MSRP.
7. Develop an interactive dashboard that supports user-driven data exploration.

5. Dataset Overview

5.1 Dataset Description

The project uses the Electric Vehicle Population dataset obtained from Kaggle.

Dataset Source:
https://www.kaggle.com/datasets/willianoliveiragibin/electric-vehicle-population

The dataset contains information about electric vehicles registered in Washington State, USA.

Each record represents an individual vehicle and includes information about its manufacturer, model, model year, vehicle type, geographical location, and selected vehicle characteristics.

5.2 Dataset Characteristics

Attribute	**Description**
Dataset	Electric Vehicle Population Data
Source	Kaggle
Geographical Scope	Washington State, USA
Number of Records	130,443
Number of Columns	17
Data Format	Structured tabular data
Visualization Tool	Tableau

5.3 Key Variables

Variable	**Description**	**Analytical Relevance**
VIN (1-10)	First ten characters of the vehicle identification number	Vehicle identification
County	County associated with the vehicle	**Geographical Context**
City	City associated with the vehicle	**Geographical Context**
State	**State of Registration**	Geographical scope
Model Year	**Vehicle model year**	Model-year distribution
Make	Vehicle manufacturer	Manufacturer comparison
Model	Specific vehicle model	Model-level comparison
Electric Vehicle Type	BEV or PHEV classification	Vehicle-type distribution
Electric Range	Recorded electric driving range	Range analysis
Base MSRP	Manufacturer-suggested retail price	Pricing analysis

These variables support the descriptive analysis and dashboard visualizations.

6. Data Preparation and Analytical Considerations

The dataset was used as the foundation for the Tableau dashboard. Relevant fields were assigned appropriate roles and aggregations to support the visualizations.

The following considerations were incorporated into the workbook:

6.1 Vehicle Record Counting

The overall vehicle KPI uses the count of DOL Vehicle ID records to represent the vehicle population.

Manufacturer and model comparisons use a distinct count of the VIN (1–10) field.

These calculations are used to summarize the vehicle population across different analytical views.

6.2 Model Year

Model Year was used as a year-based dimension to analyse the distribution of vehicle records across different model years.

The resulting visualization should be interpreted as a model-year distribution rather than a direct measure of annual sales or registration activity.

6.3 MSRP Filtering

The Average MSRP KPI excludes values below $1.

This calculation is intended to summarize average pricing among records with positive MSRP values, rather than including zero or unavailable price values in the average.

MSRP is displayed in US dollars (USD), consistent with the dataset's geographical context.

6.4 Vehicle Type Classification

The dataset contains two electric vehicle classifications:

- Battery Electric Vehicle (BEV)
- Plug-in Hybrid Electric Vehicle (PHEV)

These categories are used to compare the distribution of different vehicle types.

7. Key Performance Indicators

The dashboard includes four KPI cards that summarize the electric vehicle population.

KPI	Calculation	Purpose
Total EV Vehicles	COUNT (DOL Vehicle ID)	Represents the total number of vehicle records
Average Electric Range	AVG (Electric Range)	Summarizes the recorded electric range
Average MSRP	AVG (Base MSRP), with positive MSRP values included	Summarizes vehicle pricing among available positive values
Total EV Manufacturers	COUNTD (Make)	Represents the number of distinct manufacturers

The KPI cards provide a high-level overview of the dataset and allow users to understand its overall characteristics before examining individual visualizations.

8. Dashboard Design and Visualization

The Tableau dashboard combines summary metrics and visualizations into a single analytical interface.

Each visualization addresses a different aspect of the electric vehicle population.

8.1 KPI Cards

The dashboard includes four summary indicators:

- Total EV Vehicles
- Average Electric Range
- Average MSRP
- Total EV Manufacturers

These indicators provide a quick overview of the dataset and make important summary metrics accessible without requiring detailed exploration.

8.2 Model-Year Distribution: Overall Chart

The line chart displays the count of vehicle records across different model years.

The visualization helps identify how the representation of electric vehicles varies across model-year categories.

It provides an overview of the model-year distribution and makes it easier to compare the relative number of records associated with different years.

The chart should not be interpreted as a direct measure of annual EV sales or registrations because it uses Model Year rather than registration date.

8.3 Manufacturer Comparison

The manufacturer comparison chart displays vehicle counts across different automobile manufacturers.

This visualization makes it possible to compare the representation of manufacturers within the dataset and identify those with relatively higher vehicle counts.

A manufacturer selector allows users to highlight a selected manufacturer within the chart.

This feature improves the comparison experience by helping users focus on a particular manufacturer while retaining the broader distribution for context.

8.4 Electric Vehicle Type Distribution

The donut chart illustrates the distribution of Battery Electric Vehicles and Plug-in Hybrid Electric Vehicles.

It provides a proportional comparison between the two categories and helps users understand their relative representation within the dataset.

The visualization also supports further exploration through the electric vehicle type control.

8.5 Model-Wise Vehicle Distribution

The bubble chart displays vehicle models using circular marks.

The size of each bubble represents the vehicle count associated with the respective model.

Larger bubbles indicate models with higher vehicle counts, while smaller bubbles represent models with lower counts.

The manufacturer and model labels provide additional context for comparing individual vehicles.

This visualization helps users examine model-level distribution and identify differences in vehicle representation.

8.6 Detailed Vehicle Table

The detailed table provides a more granular view of the vehicle population.

It organizes vehicle records by manufacturer, model, and model year, allowing users to examine individual categories more closely.

The table complements the aggregated visualizations by providing additional detail for users who want to explore the dataset beyond summary charts.

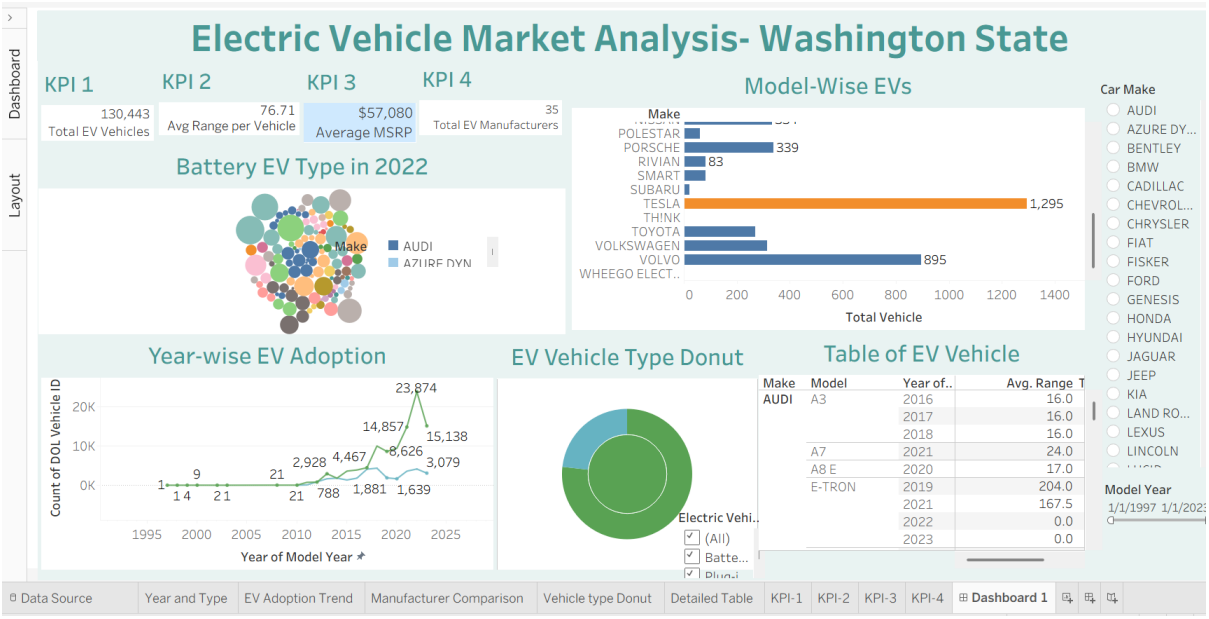


Figure 1: Interactive Tableau dashboard analysing the electric vehicle population in Washington State. The dashboard combines summary KPIs, vehicle-type distribution, manufacturer comparisons, model-year analysis, and detailed vehicle information.

9. Dashboard Interactivity

The dashboard includes interactive controls to support flexible data exploration.

9.1 Model Year Control

The Model Year control allows users to select a specific model year or a range of model years.

This enables users to examine how the displayed metrics and visualizations change when focusing on selected vehicle generations or years.

9.2 Electric Vehicle Type Control

The Electric Vehicle Type control allows users to explore the dataset by BEV and PHEV categories.

This makes it easier to examine the characteristics and distribution of different electric vehicle types.

9.3 Manufacturer Selector

The manufacturer selector allows users to highlight a selected manufacturer in the manufacturer comparison visualization.

Unlike a global manufacturer filter, this selector is intended to emphasize a manufacturer within the comparison chart.

These interactive features make the dashboard more flexible and support user-driven exploration of the data.

10. Key Analytical Observations

The dashboard supports several descriptive observations about the electric vehicle population.

10.1 Variation Across Model Years

The model-year chart shows variation in the number of vehicle records associated with different model years.

The visualization allows users to identify model-year categories with higher or lower representation and examine how the distribution changes across years.

The observed pattern reflects the composition of the available vehicle population rather than directly measuring annual sales growth.

10.2 Differences in Manufacturer Representation

The manufacturer comparison shows that vehicle counts vary across manufacturers.

The visualization provides an overview of the relative representation of different automobile manufacturers within the dataset.

It can help identify manufacturers with higher vehicle counts and provide a basis for further competitive analysis.

However, the number of records in this dataset should not automatically be interpreted as overall market share or sales volume.

10.3 Distribution of BEVs and PHEVs

The vehicle-type visualization provides a comparison between Battery Electric Vehicles and Plug-in Hybrid Electric Vehicles.

It allows users to examine the relative representation of both vehicle types and understand the composition of the dataset.

The findings are limited to the vehicles included in the dataset and do not independently establish consumer preferences across the entire automobile market.

10.4 Model-Level Distribution

The bubble chart highlights differences in vehicle counts across individual models.

The visualization makes it easier to identify models with comparatively higher representation and examine how vehicle counts vary across manufacturers and models.

This provides a useful foundation for more detailed model-level comparisons.

10.5 Vehicle Range and Pricing Characteristics

The average electric range and average MSRP KPIs provide additional information about the technical and pricing characteristics of the vehicles.

Average electric range summarizes the recorded range values, while average MSRP summarizes the available positive pricing values.

These metrics provide useful context for understanding the dataset but should not be treated as definitive measures of vehicle efficiency or affordability across the entire EV market.

11. Recommendations and Potential Further Analysis

The dashboard provides a descriptive overview of the electric vehicle population. The following areas could be explored in future analytical work.

11.1 Regional EV Distribution

The dataset contains geographical fields such as County and City.

These could be used to develop a dedicated geographical analysis to examine differences in EV representation across regions.

11.2 Pricing Segmentation

Further analysis could classify vehicles into different MSRP ranges and compare vehicle counts across price segments.

This could provide a more detailed understanding of how the recorded EV population is distributed across pricing categories.

11.3 Manufacturer and Model Comparison

Additional analysis could compare manufacturers and models across electric range, pricing, and vehicle counts.

This would provide a more detailed view of the characteristics of different vehicle categories.

11.4 Electric Range Analysis

Electric range could be examined across model years, manufacturers, and vehicle types to identify differences in recorded range characteristics.

11.5 Integration of Additional Data

Combining the dataset with additional information about charging infrastructure, vehicle sales, regional economic indicators, and government incentives could support more extensive analysis.

These recommendations represent potential directions for further investigation rather than conclusions established by the current dashboard.

12. Limitations

Several limitations should be considered when interpreting the analysis.

Geographical Scope:
The dataset is limited to Washington State and should not be generalized to the entire United States or global EV market.

Descriptive Analysis:
The project focuses on descriptive analytics and visualization. It does not establish causal relationships or predict future EV demand.

Model Year Interpretation:
The model-year chart represents vehicle distribution by model year and should not be interpreted as a direct annual sales or registration trend.

MSRP Availability:
The average MSRP calculation excludes values below $1. Consequently, the average reflects records with positive MSRP values rather than every vehicle in the dataset.

Vehicle Range Data:
The average electric range depends on the values recorded in the dataset. Missing or unavailable range information may influence how the metric should be interpreted.

External Factors:
The dataset does not independently establish the influence of factors such as charging infrastructure, government incentives, consumer preferences, or household income.

13. Post-Project Review and Opportunities for Improvement

Following the completion of the dashboard, I revisited the project to evaluate its analytical clarity, consistency, and usability. This review helped me identify several opportunities to improve the original implementation. These observations are documented separately from the completed dashboard to distinguish the original deliverable from the improvements identified during reflection.

1. Axis Scaling and Chart Readability

The model-year chart could benefit from more appropriate axis scaling. Reviewing the original visualization highlighted the importance of avoiding unnecessarily wide or negative axis ranges when displaying vehicle counts. Using a zero-based or automatically adjusted

scale where appropriate could improve readability and make differences between model years easier to interpret.

2. Clarifying Manufacturer Selection Behavior

The manufacturer selection control in the original dashboard highlights the selected manufacturer in the manufacturer comparison chart rather than filtering every dashboard visualization. This behavior could be communicated more clearly through its label or a brief instruction, helping users understand how the control affects the dashboard.

3. Consistency in Vehicle Count Measures

The dashboard uses different counting approaches in different visualizations, including counting vehicle IDs and counting distinct VIN values. A future refinement would involve validating these measures against one another and standardizing the counting logic where appropriate to ensure consistency across KPIs and charts.

4. Improving Metric Definitions

The Average Base MSRP KPI excludes records with MSRP values below $1. While this helps avoid including zero or unavailable MSRP values in the average, the dashboard or report could make this calculation more explicit so users understand which records contribute to the metric.

These observations reflect a post-project review focused on improving analytical accuracy, transparency, and user experience. They also demonstrate the importance of revisiting completed analytical work and identifying improvements beyond the initial implementation.

14. Conclusion

The Electric Vehicle Population Analysis project demonstrates the practical application of Tableau in business intelligence and descriptive analytics.

The dashboard provides a consolidated view of the recorded electric vehicle population in Washington State through KPI cards, manufacturer comparisons, model-year distribution, vehicle-type analysis, model-level comparisons, and a detailed vehicle table.

The use of interactive controls allows users to explore selected model years and vehicle types while highlighting manufacturers within the comparison chart.

By presenting the data through multiple analytical perspectives, the dashboard makes the vehicle population easier to understand and supports exploratory analysis.

Overall, the project demonstrates how business intelligence tools can transform structured datasets into an interactive and accessible analytical dashboard. It also provides a

foundation for further investigation into regional EV distribution, vehicle pricing, manufacturer comparisons, and electric range characteristics.